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Project Title: News Summarization and Sentiment Analysis Website

Relevance to Programme:

This project demonstrates practical applications of the theoretical knowledge acquired during my studies. By applying Python and NLP (Natural Language Processing) techniques for summarization and sentiment analysis, it integrates fundamental aspects of text mining, machine learning, and web development.

Abstract

The project aims to develop a website that generates summaries of news articles and performs sentiment analysis on their content. The primary objective is to provide users with concise and informative summaries while indicating the overall sentiment of the news. The methodology involves web scraping, text mining, and sentiment analysis techniques using NLP. The expected outcome is an accessible web platform that presents accurate news summaries and sentiment insights, reducing the time users spend on consuming news.

Introduction

With the rapid growth of communication technology, people obtain news from various channels, making it extremely challenging to stay efficiently updated. In industries such as politics, business, and stock market forecasting, where analyzing news sentiment is crucial for predicting trends and making informed decisions, it is especially important to quickly obtain key information from the news. This project aims to meet this need by developing a website that can condense lengthy news articles into concise summaries while also providing sentiment analysis. This will help users quickly grasp the core content of the news and understand its sentiment, enhancing their news consumption experience.

Project Requirements

Functional Requirements.

1. **Text Summarization:** Develop text summarization algorithms to generate concise

summaries of the news articles. This includes extractive summarization techniques.

2. Sentiment Analysis: Apply sentiment analysis models to determine the sentiment of news articles. This involves using existing sentiment analysis tools like VADAR.

3. User Interface: Create a user-friendly web interface to display the news summaries and sentiment scores. The interface should be intuitive, allowing users to easily navigate and access the summarized content.

Non-Functional Requirements:

1. Scalability: Ensure the system is scalable to handle a large volume of news articles. This involves optimizing the web scraping and text processing algorithms to work efficiently.

2. Accuracy: Maintain high accuracy and relevance in both the summaries and sentiment analysis. This requires thorough testing and validation of the models to ensure they produce reliable results.

Literature Review

Current text mining technologies have made significant progress, and there are mature and reliable models in the fields of text summarization and sentiment analysis.

There are three main approaches to text summarization in the field of NLP. In 2023, Karjule, Dange, Thange, et al. provided a comprehensive review of the historical development of text summarization techniques. They compared the advantages and disadvantages of extractive, abstractive, and hybrid approaches in various contexts.

Devlin, Chang, Lee, and Toutanova (2018) published a paper on the BERT model, detailing its design, and pre-training methods, and demonstrating its outstanding performance across various natural language processing tasks. In 2019, Ian Tenney, Dipanjan Das, and Ellie Pavlick mentioned BERT's potential in generative summarization and text understanding tasks, including sentiment analysis.

However, most existing solutions focus on either summarization or sentiment analysis independently. This project aims to fill this gap by developing a comprehensive website to provide both concise summaries and sentiment analysis of news articles. This dual functionality will satisfy specific needs.

Methodology

This project employs the following methodology to develop news summarization and

sentiment analysis:

1. Web Scraping: First, we will use web scraping tools to capture news articles, including headlines, article authors, publish dates, and full article content. To ensure users receive accurate data, the web scraper must precisely extract content from web pages. Common and reliable web scraping tools such as Newspaper and Scrapy will be used for this purpose.

2. Text Processing: After collecting the content, text processing is required. The articles will be segmented into sentences to facilitate subsequent summarization. We will use NLP tools like NLTK (Natural Language Toolkit) to handle the text data. These tools provide various methods for sentence segmentation, stop-word removal, and part-of-speech tagging.

3. Summarization: Summarization is one of the important functions of this project. Considering data limitations, the project will adopt extractive summarization methods such as TF-IDF, by calculating the weights of individual words to capture important vocabulary and generate summaries.

4. Sentiment Analysis: Sentiment analysis is another crucial aspect of this project. We will utilize the VADER sentiment analysis tool, which includes sentiment libraries that enable us to discern the emotional tone of articles and categorize news content into sentiment categories like positive, negative, and neutral without requiring training data.

5. Web Development: To provide a user-friendly interface for presenting summaries of news and sentiment analysis, we will develop a web application using Python Flask. Flask is a lightweight web framework that allows for the rapid development of web applications. By utilizing Flask, we can create a scalable and maintainable web application that allows users to access the above functionalities directly through their web browsers.

Through these steps, we can develop a comprehensive web application for news summarization and sentiment analysis, allowing users to grasp the content and sentiment of news articles quickly.

Project Management

The proposed timeline for the project:

July 1 - July 10: Initial Research and Planning

- Conduct a literature review
- Define project requirements.
- Set up the development environment.

Resources: Research paper

Deliverables: A specific project plan

July 11 - July 15: Web Scraping

- Test web scraping tools.

Resources: Web scraping tools, news URL

Deliverables: an initial web scraper

July 15 - July 20: Text Preprocessing

- Develop text preprocessing tasks using NLTK.

Resources: NLTK library, text processing tools.

Deliverables: Preprocessed text data module.

July 20 - August 5: Summarization module

- Generate a summarization module
- Generate a sentiment analysis module

Resources: TF-IDF, VADER

Deliverables: A module combines summarization and sentiment analysis functions.

August 5 - August 10: Web Application Development

- Develop web application using Python Flask.
- Integrate scraping, summarization, and analysis modules with the web interface.

Resources: Flask framework, front-end development tools.

Deliverables: Initial web application.

August 10 - August 15: Testing and Finalization

- Conduct thorough testing (unit, integration, user testing).
- Debug and refine the application based on feedback.
- Prepare final documentation.

Resources: Testing tools, user feedback.

Deliverables: Finalized web application, and documentation.

Conclusion

A news summarization and sentiment analysis web application addresses the need for efficient news consumption in an era of information overload. By applying NLP techniques, the project provides users with concise news summaries and sentiment insights. It demonstrates the practical application of web development, machine learning, and NLP concepts. Through careful planning, implementation, and validation, this project aspires to deliver a valuable tool that can transform the way people engage with news content.

Resources

Conroy, J. M., & O'leary, D. P. (2001). Text summarization via hidden Markov models. In Proceedings of the 24th annual international ACM SIGIR conference on Research and development in information retrieval, 406-407.

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805.

Tenney, I., Das, D., & Pavlick, E. (2019). BERT Rediscovered the Classical NLP Pipelines. Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 4593–4601.

Thange, S., Dange, J., Karjule, V., Sase, J., & Kokate, N. (2023). A Survey on Text Summarization Techniques. International Journal of Scientific and Research Publications, 13(11), 528–535.